



Inspiring Excellence

CSE422: Artificial Intelligence

Project Report

Project Title: Telco Customer Churn Prediction

Section: 9

Group no: 5

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Introduction

Customer churn is the loss of subscribers to a competing service. It is one of the costliest problems facing subscription-based businesses such as telecom providers. This project aims to predict whether an individual customer is likely to discontinue their telecom service (churn) based on their demographic profile, account information, and subscribed services.

Predicting churn in advance allows a company to take measures to prevent at-risk customers from discontinuing their service, which directly protects recurring revenue. This project also compares how different classification algorithms like Logistic Regression, Decision Tree, Neural Network etc. behave on the same real world, moderately imbalanced dataset.

Dataset Description

The dataset used is telco_customer_churn.csv, containing 7,043 customer records and 21 columns. The customerID column is a unique identifier with no predictive value and is dropped during preprocessing. The remaining 20 columns consist of 19 input features and the target variable, Churn.

This is a supervised binary classification problem, since the target variable Churn takes only two discrete values- Yes and No, rather than a continuous numeric quantity.

Feature Type	Features
Quantitative	tenure, MonthlyCharges, TotalCharges
Binary categorical	gender, SeniorCitizen, Partner, Dependents, PhoneService, PaperlessBilling
Multi-level categorical	MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaymentMethod

Sixteen of the nineteen input features are categorical (text-valued). Since none of the machine learning models used -Logistic Regression, Decision Tree, Neural Network, or K-Means can operate on text directly, categorical encoding is a required preprocessing step..

A correlation heatmap was produced over all features and the target. Since Pearson correlation requires numeric input, categorical columns were temporarily label-encoded for this step only; the encoding actually used for modeling is one-hot encoding.

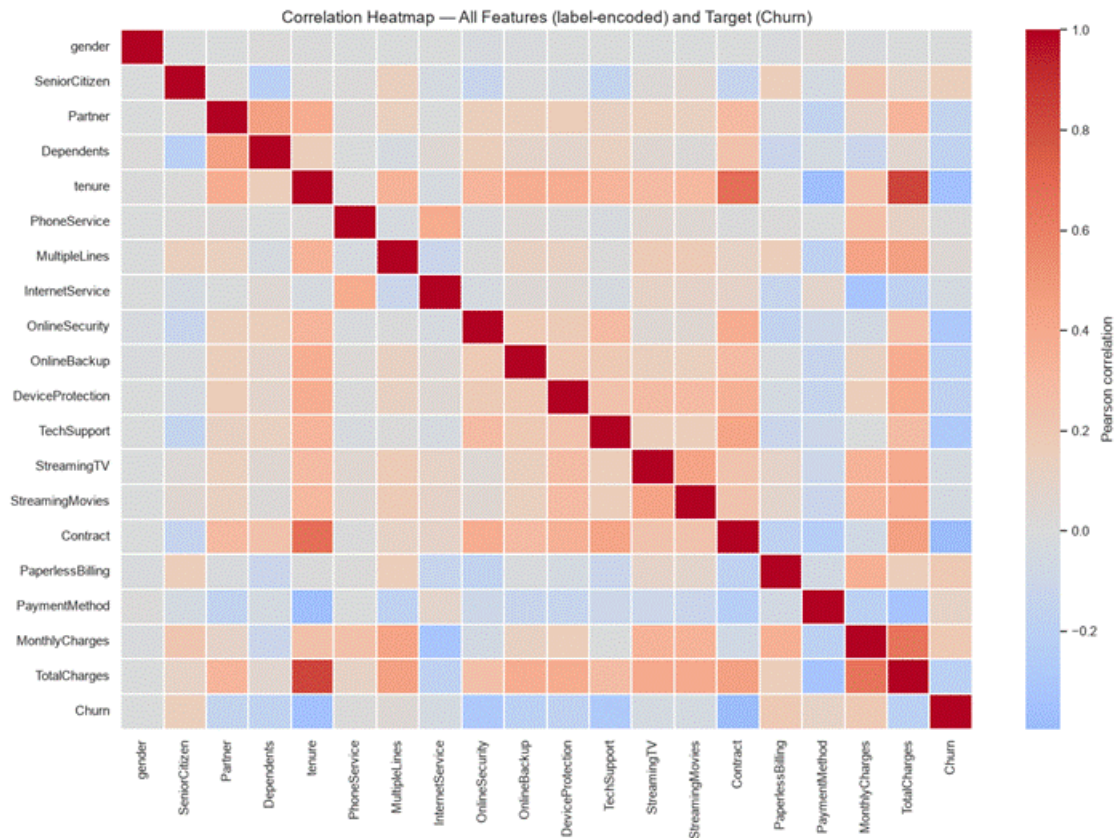


Fig. Correlation heatmap of all label-encoded features and the target variable, Churn.

Contract type and tenure show the strongest (negative) associations with churn, indicating that customers on longer commitments and longer-tenured customers are less likely to leave. OnlineSecurity and TechSupport also show a moderate negative association, while MonthlyCharges is positively correlated that is higher recurring cost is linked to a higher likelihood of churning.

Imbalanced Dataset

The target variable is moderately imbalanced: 5,174 customers (73.46%) did not churn, while 1,869 (26.54%) did which is roughly a 2.8:1 ratio. A trivial model predicting "No churn" for everyone would already reach ~73.5% accuracy while identifying zero actual churners, which is why precision, recall and AUC are used alongside accuracy, and the train/test split is stratified.

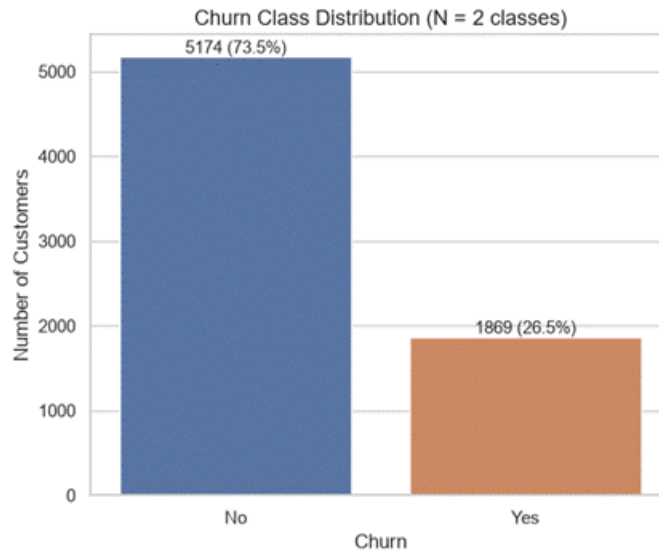


Fig. Class distribution of the target variable, Churn.

Exploratory Data Analysis

Churn rate was examined across several key categorical drivers: contract type, internet service type, and payment method. Month-to-month customers churn at 42.7% versus 11.3% (one-year) and 2.8% (two-year contracts). Fiber-optic customers churn at 41.9% versus 19.0% (DSL) and 7.4% (no internet). Customers paying by electronic check churn at 45.3%, notably higher than the 15–19% range for the other payment methods.

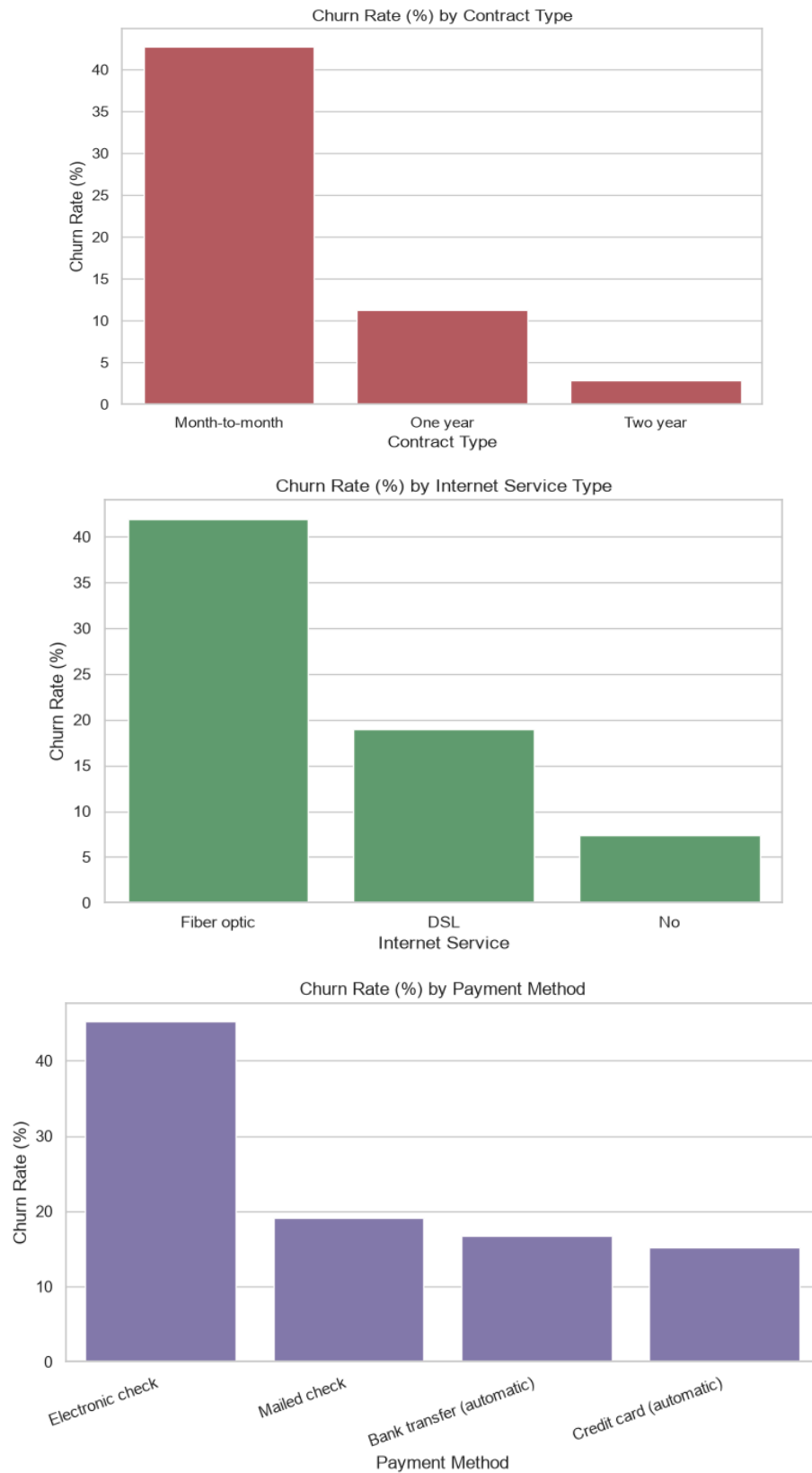


Fig. Churn rate (%) by contract type, internet service type, and payment method.

Churned customers are concentrated at low tenure and tend to have higher monthly charges than customers who stayed, consistent with the contract-length and fiber-optic findings above.

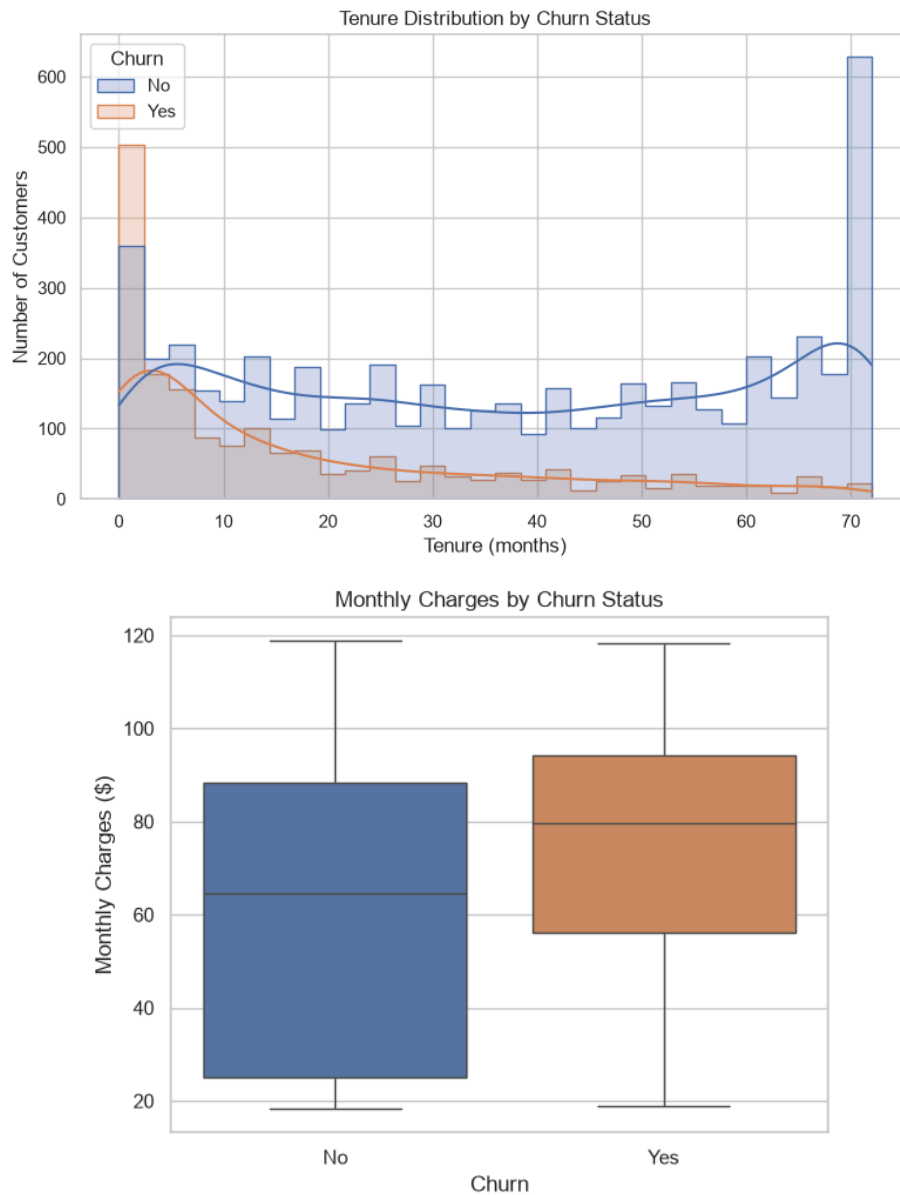


Fig. Distribution of tenure and monthly charges, split by churn status.

Senior citizens also churn at a noticeably higher rate (41.7%) than non-seniors (23.6%), despite being a minority of the customer base.

Dataset Pre-processing

Each data-quality issue is discussed together with the solution applied to it, before moving to the next. All transformations were applied to in-memory copies; the original CSV was never modified.

#	Fault	Solution
1	TotalCharges is stored as text; 11 rows contain a blank space instead of a number, so a standard null-check misses them. All 11 rows have tenure = 0 (brand-new, unbilled customers).	Converted to numeric type; the 11 blanks imputed with the domain-justified constant 0 (not a derived statistic, so no leakage risk).
2	SeniorCitizen is coded 0/1 while every other binary feature uses "No"/"Yes".	Remapped to "No"/"Yes" for consistent encoding.
3	customerID is a unique identifier with no predictive value.	Dropped before modeling.
4	16 of 19 input features are categorical (text) and unusable by any of the required models.	One-hot encoded (binary collapse for 2-level features; full one-hot for multi-level nominal features, since they have no natural order).
5	tenure, MonthlyCharges, and TotalCharges span very different numeric ranges (0–72, 18–119, up to ~8,685).	Standardized with StandardScaler (zero mean, unit variance).

To prevent data leakage, the encoder and scaler were fit only on the training split and simply applied : never re-fit to the test split, so no information from the test set influences the training representation.

Dataset Splitting

The cleaned dataset was split into training and test sets using an 80% / 20% ratio with stratified sampling on the Churn target (random_state = 42), so the 73.46% / 26.54% class is preserved in both subsets rather than left to chance.

Split	Rows	Percentage	Class balance (No / Yes)
Training set	5,634	80%	73.46% / 26.54%
Test set	1,409	20%	73.46% / 26.54%

After one-hot encoding and scaling, the 19 raw input columns expand to 40 numerical features, forming the final feature matrices used for every model.

Model Training and Testing

Three supervised classification models — Decision Tree, Logistic Regression, and a Neural Network — were trained and evaluated on identical, leak-free preprocessed data, satisfying the

requirement of a Neural Network plus at least two other models. K-Means was additionally applied as the required unsupervised component.

Decision Tree

Classifies customers via a sequence of if/else feature splits. It needs no feature scaling and is directly interpretable. Depth was limited to `max_depth = 5` to keep the tree readable and reduce overfitting. Test performance: accuracy 0.7984, precision 0.6347, recall 0.5668, AUC 0.8297.

Logistic Regression

Estimates churn probability from a linear combination of input features via the logistic function, a highly interpretable baseline. Default L2 regularization was used with `max_iter` raised to 1000 to ensure convergence on the 40 dimensional feature space. Test performance: accuracy 0.8055, precision 0.6572, recall 0.5588, AUC 0.8420 - the best result on three of the four metrics.

Neural Network (MLP)

A feed-forward network with two hidden layers (32,16 units), ReLU activation, and early stopping, capable of capturing non-linear feature interactions that a linear model cannot. Test performance: accuracy 0.7999, precision 0.6361, recall 0.5749, AUC 0.8389- the best recall among the three models.

Unsupervised Learning: K-Means Clustering

K-Means was applied using only the feature values. Candidates $K = 2$ to 10 were compared using inertia(elbow method) and average silhouette score; $K = 3$ gave the highest silhouette score (0.2751) and was selected.

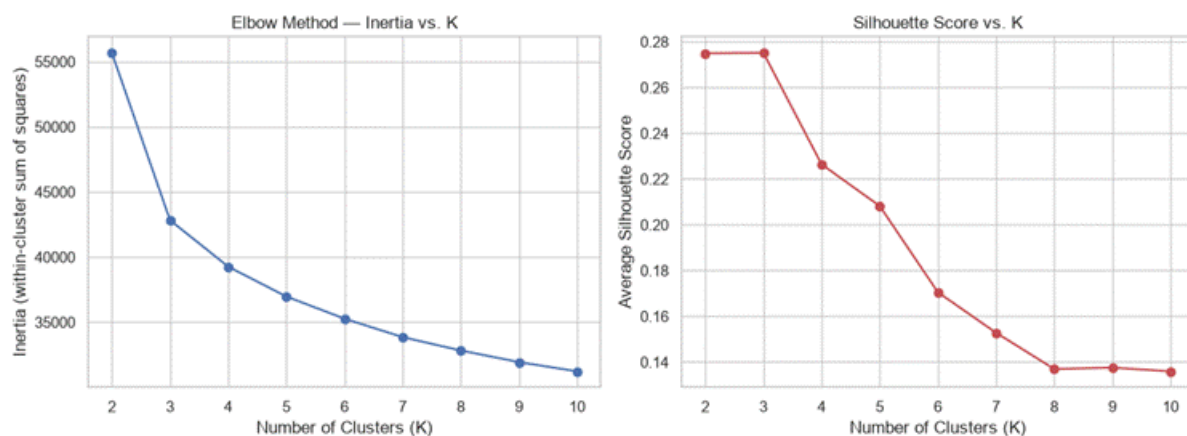


Fig. Elbow method (inertia) and average silhouette score for $K=2-10$.

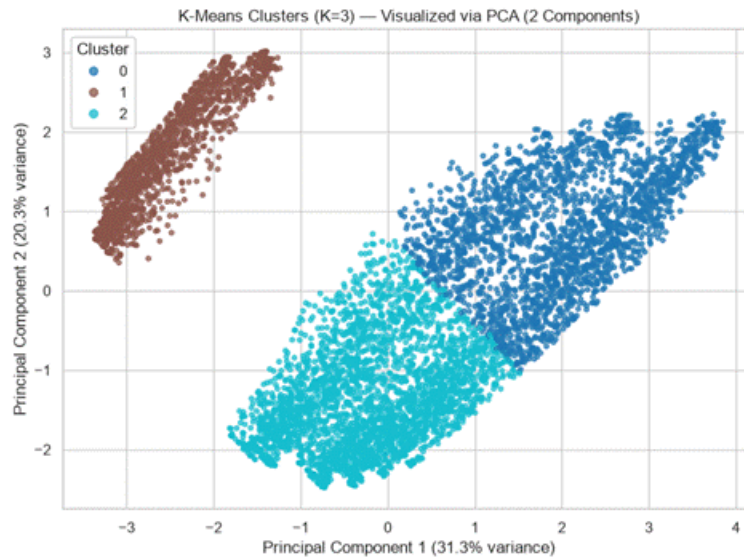


Fig. K-Means clusters (=3) visualized in two dimensions via PCA.

Churn was reintroduced afterward only to interpret the clusters, not to form them. The three clusters correspond to distinct customer segments with very different churn rates:

Attribute	Cluster 0	Cluster 1	Cluster 2
Avg. tenure	57.1 months	30.6 months	15.3 months
Avg. MonthlyCharges	\$89.12	\$21.08	\$67.91
Dominant contract	Two year	Two year	Month-to-month (87.0%)
Dominant internet	Fiber optic	No internet	Fiber optic
Churn rate (post-hoc)	14.98%	7.40%	44.08%

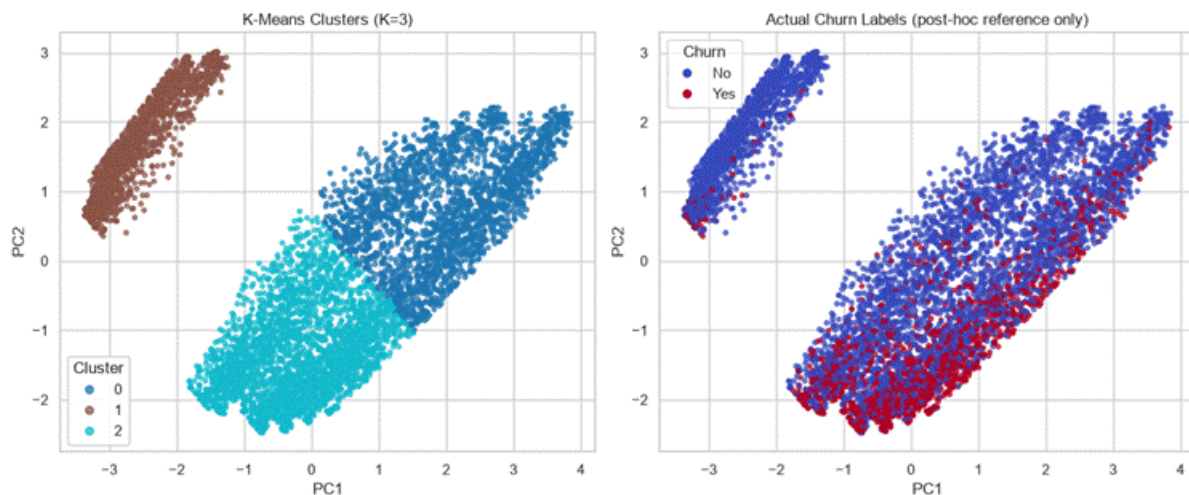


Fig. K-Means cluster assignments (left) vs. actual Churn labels (right), shown for post-hoc interpretation only.

Even without ever seeing the churn label, K-Means separated customers into segments — new, month-to-month, fiber-optic customers (Cluster 2) vs. long-tenured, contracted customers (Clusters 0–1) whose other attributes correlate strongly with actual churn behavior.

Model Selection/Comparison Analysis

The three supervised models are compared on the identical test set using accuracy, precision, recall, confusion matrices, and ROC-AUC.

Model	Accuracy	Precision	Recall	AUC
Decision Tree	0.7984	0.6347	0.5668	0.8297
Logistic Regression	0.8055	0.6572	0.5588	0.8420
Neural Network	0.7999	0.6361	0.5749	0.8389

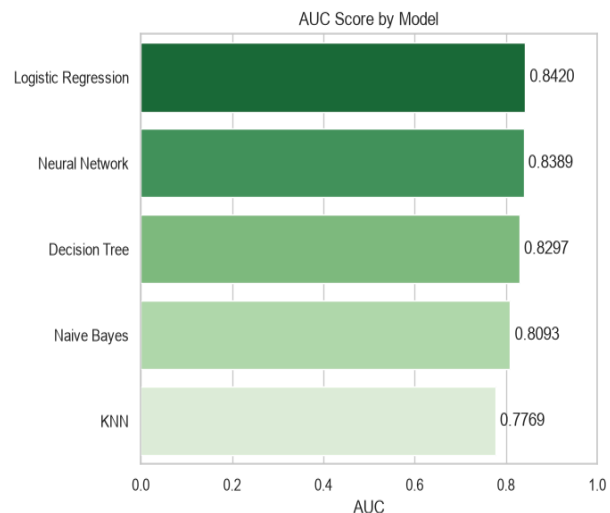
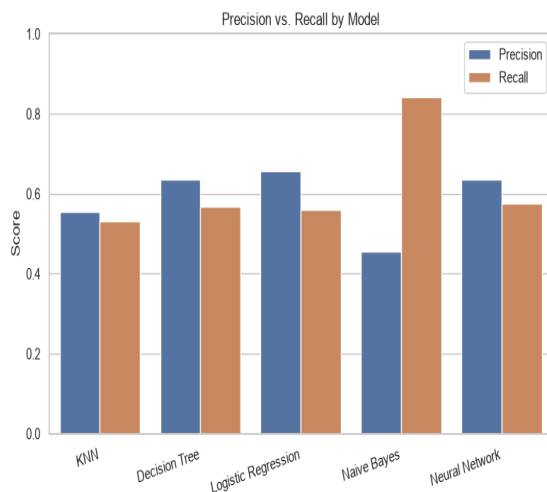
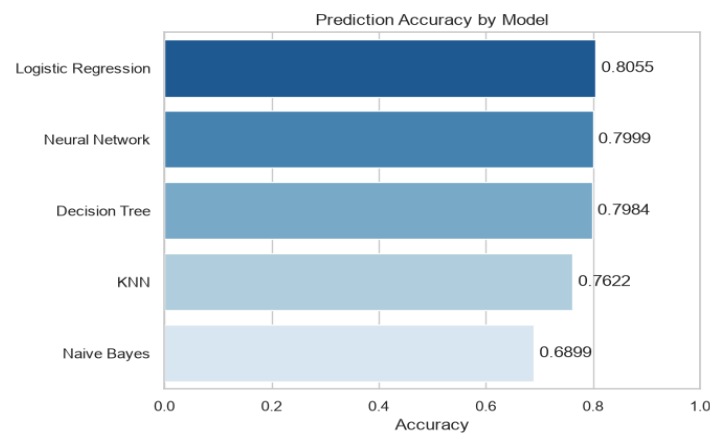


Fig Accuracy, precision/recall, and AUC comparison across all three models.

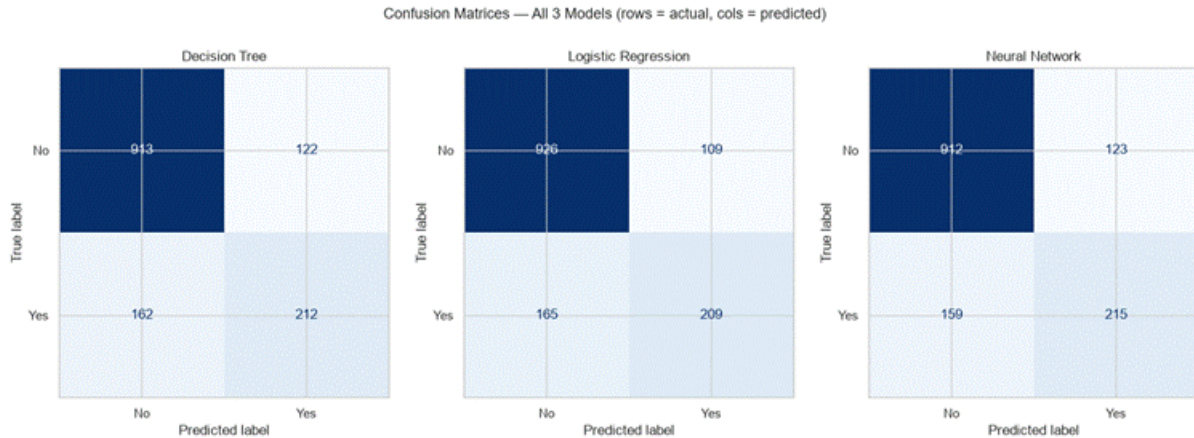


Figure 9. Confusion matrices for all three models (rows = actual class, columns = predicted class).

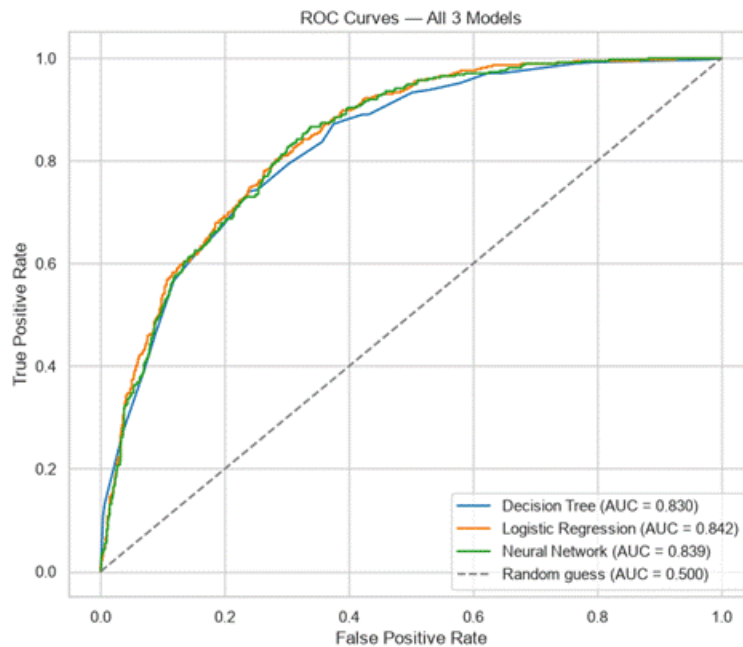


Figure 10. ROC curves for all three models, overlaid for comparison.

Logistic Regression leads on three of four metrics -accuracy, precision, and AUC , while the Neural Network holds a narrow recall advantage (0.5749 vs. 0.5588) meaning it catches slightly more actual churners. The Decision Tree trails on every metric but remains the most directly interpretable rule-based option.

Because Churn is imbalanced (73.46% / 26.54%), accuracy alone would be misleading: a model that always predicts "No churn" would score ~73.5% accuracy with 0% recall. This is why precision, recall, and AUC were tracked throughout rather than relying on accuracy in isolation.

Overall, Logistic Regression is the most defensible choice from this comparison, given its lead on three of four metrics and its interpretability for explaining churn drivers to the business,

though the Neural Network's small recall edge could matter more if the cost of missing an actual churning customer outweighs the cost of an unnecessary retention offer.

Conclusion

Churn in this dataset is not randomly distributed: it is strongly associated with contract length, tenure, internet service type, payment method, and monthly charges. Customers on month-to-month contracts, with short tenure, on fiber-optic internet or paying by electronic check consistently show much higher churn rates than their counterparts.

Among the three supervised models, Logistic Regression achieved the strongest overall performance (accuracy 0.8055, precision 0.6572, AUC 0.8420), while the Neural Network held a narrow recall lead (0.5749) and the Decision Tree trailed slightly on all metrics. The gap between models is modest overall, suggesting the churn/feature relationship is close to linear and reasonably well captured even by a simple model.

The K-Means analysis, performed without access to the churn label, still uncovered three customer segments whose churn rates differ sharply (7.40% to 44.08%), showing that meaningful churn-relevant structure exists in the data independently of the supervised labels.

Challenges included detecting the disguised missing values in TotalCharges (blank strings rather than true NaNs) and handling the moderate class imbalance, which was addressed through stratified splitting and multi-metric evaluation rather than relying on accuracy alone. Model configurations were kept intentionally simple with no extensive hyperparameter tuning, so results represent a solid baseline rather than a performance ceiling. Future work could explore systematic tuning, explicit class-imbalance handling, and ensemble methods.